

ANN Model for Energy Demand and Supply Forecasting in a Hybrid Energy Supply System

Nasser Ayoub

Faculty of Industrial Education,
Helwan University,
Cairo, Egypt.

e-mail: n_ayoub@techedu.helwan.edu.eg

Farayi Musharavati

Department of Mechanical and Industrial Engineering,
College of Engineering, Qatar University,
Doha, Qatar.

e-mail: farayi@qu.edu.qa

Shaligram Pokharel

Department of Mechanical and Industrial Engineering,
College of Engineering, Qatar University,
Doha, Qatar.

e-mail: shaligram@qu.edu.qa

Hossam A. Gabbar

Faculty of Energy Systems and Nuclear Science,
University of Ontario Institute of Technology,
Ontario, Canada.

e-mail: Hossam.Gaber@uoit.ca

Abstract—This paper presents short term demand and supply forecasting model for a microgrid supply system used to secure the electricity demands of a commercial building, using one year demand data collected in hourly base. One-year renewable-based Micro-grid electricity supply data were produced by simulating its sub-systems (wind and PV supply systems). The Artificial Neural Network, ANN, forecasting models are built on predicting generation capacity and load demands in the next 24 hours. The ANN model presented here is a micro-level supply and demand forecasting model that links the decision making with the performance measures. To sustain the model results, the daily weather forecasts supplied by local authorities, are incorporated in our model. The models validity were tested by calculating the Mean Absolute Percent Error for the forecasted data. The ANN models' applicability and performance were tested in a case study for forecasting the demands of a hotel building and the supply potential of its microgrid supply sub-system. The building demands are assumed to be supplied by a hybrid supply system of 20% renewable-based Micro grid (10% Wind and 10% Photovoltaic) and 80% from electricity grid.

Keywords- energy conservation; forecasting; neural network; commercial; hybrid supply system

I. INTRODUCTION

Small scale and microgrid supply systems are used by customers for many reasons such as countering the unreliability in conventional electricity supply, enhancing the share of green electricity source, or for electricity supply to rural areas. As supply from the renewable system is not continuous, it affects the quality of electricity supply to the end-user. Therefore, in order to achieve energy supply stability and better operation performance, it is important to forecast the energy demands collaboratively. There are three main forecasting categories depending on the forecasting period covered; Short Term Forecasting (STF), Medium Term Forecasting (MTF), and Long-Term Forecasting (LTF). In general, STF focus on time range from 24 hours to

one week; MTF from one week up to one year, and; LTF focus on longer time frame than the other two types [1].

Energy consumption forecasting for energy carriers such as electricity, natural gas and oil, has been carried out by many researches for GCC countries using several statistical and AI models [2-4]. However, in spite of the substantial amount of energy consumed in different buildings' types, the energy consumption forecasting in buildings has not been investigated by the researchers in GCC countries. There are studies providing robust and effective forecasting models either for energy supply or energy demands, for example [5-8]. In the supply side, renewable energy supply forecasting is done at the macro level, where researchers have focused on large size photovoltaic plants and, to a less extent, wind electricity supply [9-15]. Also, there are several studies on forecasting natural gas consumption for gas load management by gas distribution system operators [3, 16]. The energy demand forecasting in buildings is also popular in the literature, for example [1, 7, 8, 17], but it is less extensively done in the supply side [18, 19]. This paper is focusing on STF for both energy demands and supply in buildings. In the next section, we are presenting the state of the art of the most popular forecasting models in the area of building energy supply/ demand forecasting including intelligent forecasting models such as Artificial Neural Networks, ANN, Support Vector Machine, SVM, and other statistical models such as Bagged Regression Trees models.

A. Demand Side Forecasting Models

The hourly energy consumption in buildings was forecasted, by Gonzalez, using an artificial neural network (ANN) model that feeds back a part of its outputs [20]. In this work, the numbers of neurons, the optimal data window size and the training algorithm parameters have not been deeply analyzed. A neural network-based model for electricity load forecasting 24 hours ahead of time at the building-level has been presented in [17]. The k-means clustering method is used to identify different operation values based on locations and times of day. The model is

applied in two stages namely; parameter selection phase (executed every few weeks or months) and forecasting phase (performed on a daily basis) to determine the load for the following day. The electricity energy consumption prediction has been presented in [21] by using three forecasting models (regression analysis, decision tree and neural networks). The authors claimed that the neural networks model outperformed other models when the functional form is nonlinear and when there is noise in the data.

B. Supply Side Forecasting Models

A forecast day information-based neural network approach for estimating 24-hour-ahead electricity production of a photovoltaic system is presented [9]. The model uses a similar day selection algorithm to improve forecast accuracy in different weather types. The forecasting results showed that the suggested approach has a high forecasting accuracy and efficiency of the power output. Another NN forecasting model for power output of PV grid connected system without measuring solar radiation is also presented. The model used a set of calculated solar radiation data and weather forecasting data to estimate the electricity supply potential. The proposed model has 14 inputs including the solar radiation, by considering next day's highest temperature, lowest temperature and cloudy conditions [22]. An online 24-h solar power forecasting using radial basis function neural network (RBFN) based on weather type classification is also presented. The model inputs are the mean daily solar irradiance, the mean daily relative humidity, the mean daily wind speed, the daily air temperature and the mean daily power output. The output is represented by the 24 h ahead of power output of the PV system [23].

II. DEMANDS AND SUPPLY SYSTEMS

A. Building Model

The building considered to be supplied by the hybrid energy supply system is a hotel that is modeled assuming the metrological condition of Qatar. The total floor area of the hotel is 4448 square meters. It is a nine stories establishment above the ground level, facing the north.

The hotel is assumed to be opened at all times. The energy-consuming equipment have to be switched on and off according to the time of the day. Weather conditions. Electricity is considered as the primary energy source to power HVAC, lighting, elevators, and all equipment used by the hotel occupants. Gas is used for cooking and water boiling. The building is divided into eight thermal zones as described in [24]. The hybrid supply system is explained in the following section.

B. Microgrid Power Supply System

The investigated micro grid power supply system, shown in Fig. 1, consists of the hotel load profile representing the hotel electricity consumption; renewable electricity supply system of 26 polycrystalline silicon PV cells of 4 KWp rated electric power each with a total area of

918 m² and 5 wind turbines each of 100 KWp. The rest of the power load is supplied by conventional electricity through the grid (Qatar uses 100% of NG to generate electricity). The life cycle for PV panels and wind turbines are explained in detailed in [25] and [26, 27]. The PV panels and wind turbines are connected to 4 Vanadium redox storage batteries with a storage capacity of 0.45 MWh of electrical energy. The batteries are connected to the hotel through power inverters. The electricity grid and inverters are supplying the electricity to the hotel through power controller. The annual electricity demands, calculated by simulating the hotel building, in EnergyPlus using the climate data of Qatar, is 2953 MWh. The life cycle of the microgrid components is assumed to be 25 years for all its components.

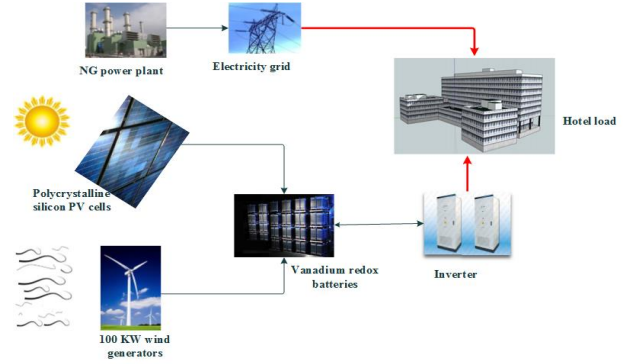


Figure 1. Schematic representation of the microgrid energy supply system for the hotel.

III. MICROGRID POWER GENERATION MODELS

A. Wind Power Generation Model

There are two models used to extrapolate wind speeds to greater heights, namely: the log law and the power law. Both models have been shown to perform equally in shear speed prediction on average [15]. The wind data are collected from Qatar International Airport station in Doha, 25.25N- 51.57E at 10 m above the ground elevation. The average wind speed, v_1 , of the measured data is 4.4 m/s. Modern wind turbines have a hub height of 50 meters and above and wind speed increases with height, v_2 , as approximated in the following log law model:

$$v_2 = v_1 \cdot \frac{\ln(\frac{h_2}{r})}{\ln(\frac{h_1}{r})}$$

where r is the roughness factor and assumed to be 0.02 m, which is typical for airport areas with buildings and trees [1]. Taking $h_1 = 10$ m and $h_2 = 60$ m, we obtain a correction factor equal to 1.289 for the original wind data. The average yearly wind speed then becomes 5.67 m/s.

A sequential plot of the wind speed is shown in Fig. 2. We see that the wind speed has moderate pattern along the year. This could have a negative effect on the building's energy balance since the electricity demands vary between

winter and summer. So, it may be better to use wind generator for supplying electricity to non-seasonal consuming equipment such as lighting systems.

$$P_T = 1/2 * \rho * A * v^3 * C_p$$

where v wind speed, ρ air density and it is assumed to be 1.255 kg/m³, C_p is power coefficient, which is the ratio of power extracted by the turbine to the total contained in the wind resource and it is assumed to be the maximal possible 0.59 or what is called Betz Limit, A is the rotor swept area.

Based on the previous equations, parameters and specification, given in Table I, the overall power supply from wind was calculated, as shown in Fig. 3.

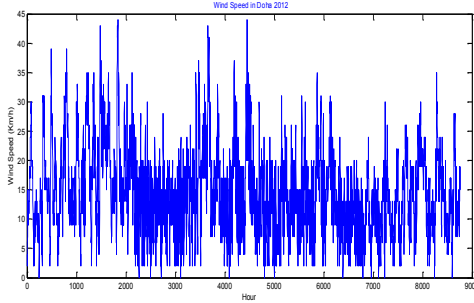


Figure 2. Wind speed profile for Doha city measured at the International Airport Station.

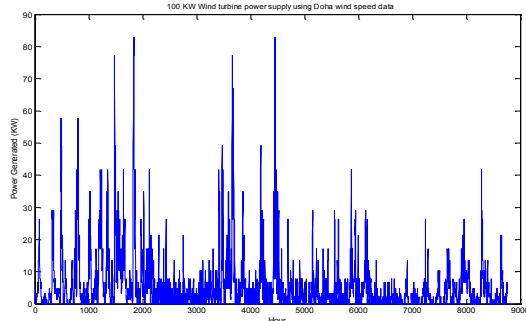


Figure 3. The 100 KW Wind turbine power supply using Doha wind speed data.

B. PV Power Generation Model

TABLE I. WIND TURBINE SPECIFICATIONS

Wind System	Value
Capacity rating	100 KW
Rotor diameter	13 m
Swept area	346 m ²
Hub height	32 m
Number of blades	3 glass fibre blades
Weather data	Qatar
The roughness factor	0.02

Cut-in wind speed	1.5 m/s
Rated wind speed	10 m/s
Cut-out wind speed	16 m/s

The daily level of horizontal irradiation in Qatar ranges from 1950 W/m²/day in winter time to 2230 W/m²/day in summer. Different irradiation levels lay over this range along the year. The output photovoltaic electricity generation depends on panel area, performance ability and durability of the panels. In this study, panel size is 144 cm², the installation lifetime of 25 years, an efficiency of 13% and a performance ratio of 0.75 are used. The PV generation system considered in this study consists of 26 polycrystalline silicon PV cells of 4 KWp rated electric power each with a total area of 918 m². Degradation due to ageing of the system was also considered as it results in a linear reduction in performance over its useful lifetime. For example, the degradation factor effectively reduces the average performance over the system lifetime by 20% [28]. The effect of dust accumulation is also considered. The reported power losses due to dust accumulation on panels are 31% to 35% on the maximum solar intensity.

The total electricity output of the PV system is calculated from the following equation:

$$P_{PV} = A \cdot \eta \cdot R_p \cdot L \cdot d \cdot I_j$$

where K is the total electricity output from the panel (Wh); A is area of the panel (m²); η is panel efficiency (%); R_p is performance ratio; d is average degradation factor and I_j is annual irradiance at installation site (Wh/m²). The performance data of the PV cells are given in Table II and their hourly output power is shown in Fig. 4.

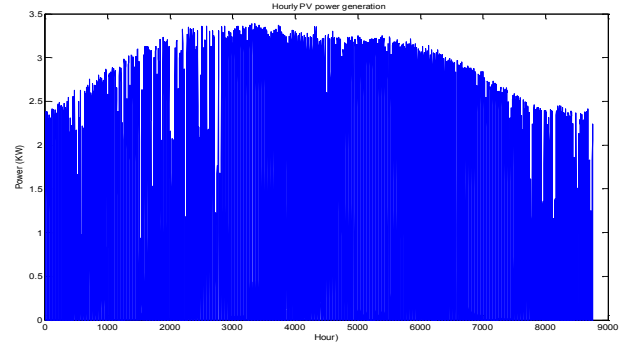


Figure 4. The 4 KW PV cells' hourly output power using Doha Irradiation Data.

TABLE II. HOURLY PERFORMANCE DATA OF PV SYSTEM ARRAYS

PV System	Value
Location	Doha
Elevation (m)	27
DC System Size (kW)	4
Module Type	Thin Film
Array Type	1-Axis Tracking
Array Tilt (deg)	0
Array Azimuth (deg)	180
System Losses	14.08

Invert Efficiency	96%
DC to AC Size Ratio	1.1
Ground Coverage Ratio	40%

IV. NEURAL NETWORK MODEL

Artificial Neural Networks, ANN, forecasting is one of the most, if not the most, popular modeling tools for energy consumption forecasting. Various types of neural networks can be used for prediction, such as back propagation, ART, Marks network and others. In this research we have focused on back-propagation ANN. The model aims at finding the most suitable structure for one day ahead forecasting of electricity consumption and electricity generation from renewable sources namely; wind and photovoltaic.

TABLE III. CONTROLLABLE AND UN-CONTROLLABLE PARAMETERS OF SUPPLY/ DEMAND SYSTEMS

Parameter	Description	Unit
Controllable parameters	Chilled water coil valve position	%Open
	Supply air humidity	% RH
	Mixed air temperature	°F
	Chilled water coil entering water temperature	°F
Uncontrollable parameters	Return air fan speed	CFM
	Outside air humidity	°F
	Outside air temperature	% RH
	Outside air CO2 concentration	PPM
	Infrared radiation	W/m ²
	Solar normal flux	W/m ²
	Solar beam	W/m ²
	Barometric pressure (normalized to sea level)	kpa
	Outside wind velocity	Km/h
	Outside wind direction	°C
Target controllable parameters	Building energy consumption	KW
	Supply system generated power	

The short-term forecast of 24 hours ahead supply/ consumption in a building via ANN includes training the network on a set of its historical data and a group of controllable (supply/ demand) and uncontrollable predictors (Table III) in a selected horizon of time. The identification of the controllable & un-controllable parameters and eliminating the redundant or irrelevant parameters is vital in the forecasting model development to improve the comprehensibility, scalability, and perhaps, accuracy of the resulting model [22]. In this work, the controllable and un-controllable parameters of the forecasting targets, either for supply and demand, were identified. Table III shows some of these parameters for energy supply/ demand systems in buildings. The target parameters of the forecasting processes are electricity demands and power generation. Using the trained network model, we can estimate the supply/ demand quantities.

The construction of the neural networks model, followed here, is a consequential adaptive process to minimize the prediction errors where the Mean Absolute Percentage Error (MAPE) is regarded as the cost function. The MAPE is

considered a standard for examining the quality of the prediction models of demand/ supply in a large number of literatures consulted. For each supply/ demand model, fifteen networks are trained with the hidden neurons varying from 20 to 30. The best network with the minimum MAPE is selected as the optimal model. In the proposed ANN model, we have created several timing predictors to identify the specific location of similar values within the supply/ demand time series as the one we are forecasting. Those predictors are hour of the day, same hour load/supply pre day, day of week, same hour load/supply every week, previous day same hour, previous week same hour, day of the week and month of the year.

The general forecasting procedure in this study is as follows:

1. Importing the historical hourly Supply/ Demand data that include controllable as well as uncontrollable parameters.
2. If the building follows a typical holiday schedule the list of holidays should be loaded to be considered in the predictors
3. Select the forecasting horizon by splitting the dataset to training and test sets. The weather data sets and consumption/ supply simulations of year 2012 are used in this study.
4. Build the load forecasting model for day-ahead load forecasting.
5. Run the model and save trained trees for future use.
6. Quantify the model performance of the forecaster using mean absolute percent error (MAPE).

V. RESULTS AND DISCUSSION

The proposed ANN model has been applied to the two electricity supply microgrid sub-systems, in addition to the building demands for forecasting of demands/ supply. The results are shown in Figures 5-7 for the demands and two supply systems respectively. By changing different parameters of the neural network architecture (e.g., number of hidden layers, number of neurons per hidden layer) we could identify the suitable neural network structure which performs the best for each controllable parameters of the building throughout the parameter selection stage of the system modeling.

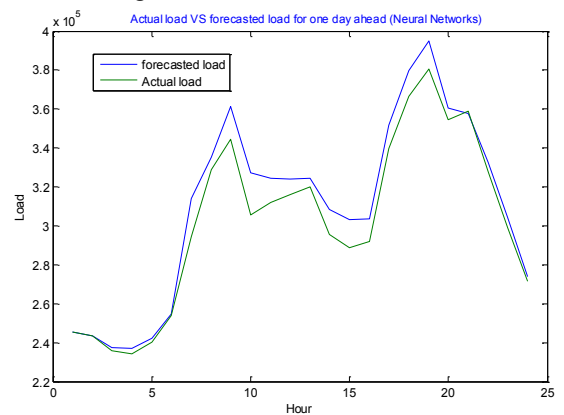


Figure 5. Load forecast results using ANN model.

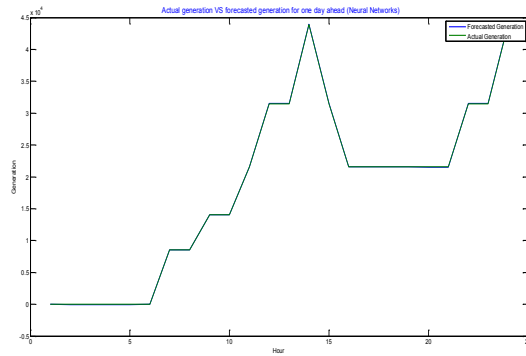


Figure 6. Wind power supply forecast results using ANN model

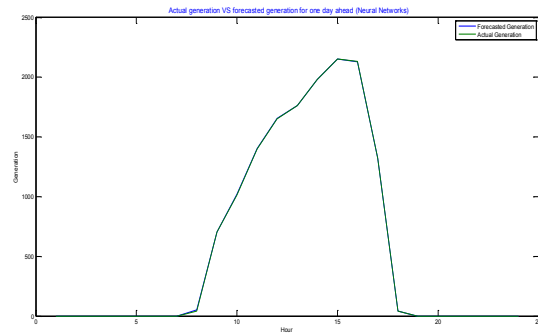


Figure 7. Photovoltaic supply forecast results using ANN Model

The electricity demands forecasting results exhibited by Fig. 5 shows the high predictability of the proposed model. Using the ANN model without predictors yielded MAPE of 2.12. When adding the uncontrollable parameters as predictors the forecasting error calculated by ANN was greatly reduced to 1.56%. The results of the forecasted wind power generation shown in Fig. 6. As seen in Fig. 3, the wind generation data are highly nonlinear exhibiting a typical ANN forecasting environment. The ANN model outperformed the demand forecasting accuracy and closely predicted the wind generation with MAPE of 0.186% that shows the high responsiveness of the proposed model in this case. The solar radiation data, used in forecasting PV power generated, are discrete in nature as they become zero at night and increase as the sunrise before they decrease again going towards the sun set. The actual photovoltaic electricity generation behaviour versus the forecasted generation are presented in Fig. 7. As seen in the figure, the ANN model has performed well in forecasting the PV power generation with an MAPE of 2.165%.

The results show that a simple NN structure is sufficient for most cases as in some cases more complex NNs gave poorer forecasting accuracy. This is desired because it leads to quicker processing intervals that speeds up the overall forecasting process and improves accuracy. Furthermore, the uncontrollable parameters influencing electricity demands/ supply tend to differ with the forecasted building zone and forecasted period; hence a further analysis is required to identify the exact influential parameters related to each forecasting session. In addition, comparing the

proposed model to the similar models developed should be thoroughly analyzed.

ACKNOWLEDGMENT

This research was made possible by a NPRP award NPRP 5 – 209 – 2 - 071 from the Qatar National Research Fund (a member of The Qatar Foundation). The statements made herein are solely the responsibility of the authors.

REFERENCES

- [1] A. S. Ahmad, M. Y. Hassan, M. P. Abdullah, H. A. Rahman, F. Hussin, H. Abdullah, et al., "A review on applications of ANN and SVM for building electrical energy consumption forecasting," *Renewable and Sustainable Energy Reviews*, vol. 33, pp. 102-109, 5/ 2014.
- [2] R. E. Abdel-Aal, "Univariate modeling and forecasting of monthly energy demand time series using abductive and neural networks," *Computers & Industrial Engineering*, vol. 54, pp. 903-917, 5/ 2008.
- [3] A. Azadeh, S. M. Asadzadeh, M. Saberi, V. Nadimi, A. Tajvidi, and M. Sheikhalishahi, "A Neuro-fuzzy-stochastic frontier analysis approach for long-term natural gas consumption forecasting and behavior analysis: The cases of Bahrain, Saudi Arabia, Syria, and UAE," *Applied Energy*, vol. 88, pp. 3850-3859, 11/ 2011.
- [4] R. E. Abdel-Aal, A. Z. Al-Garni, and Y. N. Al-Nassar, "Modelling and forecasting monthly electric energy consumption in eastern Saudi Arabia using abductive networks," *Energy*, vol. 22, pp. 911-921, 9/ 1997.
- [5] S. Jie, L. Wei-Jen, L. Yongqian, Y. Yongping, and P. Wang, "Forecasting power output of photovoltaic system based on weather classification and support vector machine," in *Industry Applications Society Annual Meeting (IAS)*, 2011 IEEE, 2011, pp. 1-6.
- [6] H. T. C. Pedro and C. F. M. Coimbra, "Assessment of forecasting techniques for solar power production with no exogenous inputs," *Solar Energy*, vol. 86, pp. 2017-2028, 7/ 2012.
- [7] Y. Jiang, Y. Yao, S. Deng, and Z. Ma, "Applying grey forecasting to predicting the operating energy performance of air cooled water chillers," *International Journal of Refrigeration*, vol. 27, pp. 385-392, 6/ 2004.
- [8] A. H. Neto and F. A. S. Fiorelli, "Comparison between detailed model simulation and artificial neural network for forecasting building energy consumption," *Energy and Buildings*, vol. 40, pp. 2169-2176, 2008.
- [9] M. Ding, L. Wang, and R. Bi, "An ANN-based Approach for Forecasting the Power Output of Photovoltaic System," *Procedia Environmental Sciences*, vol. 11, Part C, pp. 1308-1315, 2011.
- [10] C. Monteiro, T. Santos, L. Fernandez-Jimenez, I. Ramirez-Rosado, and M. Terreros-Olarte, "Short-Term Power Forecasting Model for Photovoltaic Plants Based on Historical Similarity," *Energies*, vol. 6, pp. 2624-2643, 2013.
- [11] G. Schubert, "Modeling hourly electricity generation from PV and wind plants in Europe," in *European Energy Market (EEM)*, 2012 9th International Conference on the, 2012, pp. 1-7.
- [12] E. İzgi, A. Öztopal, B. Yerli, M. K. Kaymak, and A. D. Şahin, "Short-mid-term solar power prediction by using artificial neural networks," *Solar Energy*, vol. 86, pp. 725-733, 2/ 2012.
- [13] T. R. Ayodele, A. A. Jimoh, J. L. Munda, and J. T. Agee, "Wind distribution and capacity factor estimation for wind turbines in the coastal region of South Africa," *Energy Conversion and Management*, vol. 64, pp. 614-625, 12/ 2012.
- [14] M. Korpås, "Distributed Energy Systems with Wind Power and Energy Storage," Doctoral, Department of Electrical Power Engineering, Norwegian University of Science and Technology Trondheim, 2004.
- [15] M. L. Kubik, P. J. Coker, and C. Hunt, "Using meteorological wind data to estimate turbine generation output: a sensitivity analysis," in

- World Renewable Energy Congress 2011 Sweden - linköping 2011, pp. 4074-4081.
- [16] P. Musilek, Pelika, x, E. n, T. Brabec, and M. Simunek, "Recurrent Neural Network Based Gating for Natural Gas Load Prediction System," in *Neural Networks, 2006. IJCNN '06. International Joint Conference on, 2006*, pp. 3736-3741.
 - [17] J. G. Jetcheva, M. Majidpour, and W.-P. Chen, "Neural network model ensembles for building-level electricity load forecasts," *Energy and Buildings*, vol. 84, pp. 214-223, 12/ 2014.
 - [18] Y. Iwafune, Y. Yagita, T. Ikegami, and K. Ogimoto, "Short-term forecasting of residential building load for distributed energy management," in *Energy Conference (ENERGYCON), 2014 IEEE International, 2014*, pp. 1197-1204.
 - [19] Y. Xiaohui, C. Haisheng, Z. Xuehui, and T. Chunqing, "Energy storage sizing for office buildings based on short-term load forecasting," in *Information and Automation for Sustainability (ICIAfS), 2012 IEEE 6th International Conference on, 2012*, pp. 290-295.
 - [20] P. A. González and J. M. Zamarreño, "Prediction of hourly energy consumption in buildings based on a feedback artificial neural network," *Energy and Buildings*, vol. 37, pp. 595-601, 6/ 2005.
 - [21] G. K. F. Tso and K. K. W. Yau, "Predicting electricity energy consumption: A comparison of regression analysis, decision tree and neural networks," *Energy*, vol. 32, pp. 1761-1768, 9/ 2007.
 - [22] C. Chupong and B. Plangklang, "Forecasting power output of PV grid connected system in Thailand without using solar radiation measurement," *Energy Procedia*, vol. 9, pp. 230-237, / 2011.
 - [23] C. Chen, S. Duan, T. Cai, and B. Liu, "Online 24-h solar power forecasting based on weather type classification using artificial neural network," *Solar Energy*, vol. 85, pp. 2856-2870, 11/ 2011.
 - [24] N. Ayoub, F. Musharavati, S. Pokharel, and H. A. Gabbar, "Energy consumption and conservation practices in Qatar—A case study of a hotel building," *Energy and Buildings*, vol. 84, pp. 55-69, 12/ 2014.
 - [25] A. Stoppato, "Life cycle assessment of photovoltaic electricity generation," *Energy*, vol. 33, pp. 224-232, 2/ 2008.
 - [26] L. Schleisner, "Life cycle assessment of a wind farm and related externalities," *Renewable Energy*, vol. 20, pp. 279-288, 7/1/ 2000.
 - [27] M. R. Kabir, B. Rooke, G. D. M. Dassanayake, and B. A. Fleck, "Comparative life cycle energy, emission, and economic analysis of 100 kW nameplate wind power generation," *Renewable Energy*, vol. 37, pp. 133-141, 1/ 2012.
 - [28] R. Bush, D. A. Jacques, K. Scott, and J. Barrett, "The carbon payback of micro-generation: An integrated hybrid input–output approach," *Applied Energy*, vol. 119, pp. 85-98, 4/15/ 2014.